

22.1 Features

- 32-bit counter running from PCLK. Counter can be free-running, or be reset by a generated interrupt.
- 32-bit compare value.
- 32-bit compare mask. An interrupt is generated when the counter value equals the compare value, after masking. This allows for combinations not possible with a simple compare.

22.2 Description

The Repetitive Interrupt Timer provides a versatile means of generating interrupts at specified time intervals, without using a standard timer. It is intended for repeating interrupts that aren't related to Operating System interrupts. However, it could be used as an alternative to the Cortex-M3 System Tick Timer ([Section 23.1](#)) if there are different system requirements.

22.3 Register description

Table 434. Repetitive Interrupt Timer register map

Name	Description	Access	Reset value ^[1]	Address
RICOMPVAL	Compare register	R/W	0xFFFF FFFF	0x400B 0000
RIMASK	Mask register. This register holds the 32-bit mask value.	R/W	0	0x400B 0004
RICTRL	Control register.	R/W	0xC	0x400B 0008
RICOUNTER	32-bit counter	R/W	0	0x400B 000C

[1] Reset Value reflects the data stored in used bits only. It does not include content of reserved bits.

22.3.1 RI Compare Value register (RICOMPVAL - 0x400B 0000)

Table 435. RI Compare Value register (RICOMPVAL - address 0x400B 0000) bit description

Bit	Symbol	Description	Reset value
31:0	RICOMP	Compare register. Holds the compare value which is compared to the counter.	0xFFFF FFFF

22.3.2 RI Mask register (RIMASK - 0x400B 0004)

Table 436. RI Mask register (RIMASK - address 0x400B 0004) bit description

Bit	Symbol	Description	Reset value
31:0	RIMASK	Mask register. This register holds the 32-bit mask value. A one written to any bit overrides the result of the comparison for the corresponding bit of the counter and compare register (causes the comparison of the register bits to be always true).	0

22.3.3 RI Control register (RICTRL - 0x400B 0008)

Table 437. RI Control register (RICTRL - address 0x400B 0008) bit description

Bit	Symbol	Value	Description	Reset value
0	RITINT		Interrupt flag	0
		1	This bit is set to 1 by hardware whenever the counter value equals the masked compare value specified by the contents of RICOMPVAL and RIMASK registers. Writing a 1 to this bit will clear it to 0. Writing a 0 has no effect.	
		0	The counter value does not equal the masked compare value.	
1	RITENCLR		Timer enable clear	0
		1	The timer will be cleared to 0 whenever the counter value equals the masked compare value specified by the contents of RICOMPVAL and RIMASK registers. This will occur on the same clock that sets the interrupt flag.	
		0	The timer will not be cleared to 0.	
2	RITENBR		Timer enable for debug	1
		1	The timer is halted when the processor is halted for debugging.	
		0	Debug has no effect on the timer operation.	
3	RITEN		Timer enable.	1
		1	Timer enabled. Remark: This can be overruled by a debug halt if enabled in bit 2.	
		0	Timer disabled.	
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

22.3.4 RI Counter register (RICOUNTER - 0x400B 000C)

Table 438. RI Counter register (RICOUNTER - address 0x400B 000C) bit description

Bit	Symbol	Description	Reset value
31:0	RICOUNTER	32-bit up counter. Counts continuously unless RITEN bit in RICTRL register is cleared or debug mode is entered (if enabled by the RITNEBR bit in RICTRL). Can be loaded to any value in software.	0

22.4 RI timer operation

Following reset, the counter begins counting up from 00000000h. Whenever the counter value equals the value programmed into the RICOMPVAL register the interrupt flag will be set. Any bit or combination of bits can be removed from this comparison (i.e. forced to compare) by writing a '1' to the corresponding bit(s) in the RIMASK register. If the enable_clr bit is low (default state), a valid comparison ONLY causes the interrupt flag to be set. It has no effect on the count sequence. Counting continues as usual. When the counter reaches FFFFFFFFh it rolls-over to 00000000h on the next clock and continues counting. If the enable_clr bit is set to '1' a valid comparison will also cause the counter to be reset to zero. Counting will resume from there on the next clock edge.

Counting can be halted in software by writing a '0' to the Enable_Timer bit - RICTRL(3). Counting will also be halted when the processor is halted for debugging provided the Enable_Break bit – RICTRL(2) is set. Both the Enable_Timer and Enable_Break bits are set on reset.

The interrupt flag can be cleared in software by writing a '1' to the Interrupt bit – RICTRL(0).

Software can load the counter to any value at any time by writing to RICOUNTER.

The counter (RICOUNTER), RCOMPVAL register, RIMASK register and RICTRL register can all be read by software at any time.

